4. Equilibrium Calculations

Our goal in this section is to develop a general strategy for dealing quantitatively with the equilibria of weak acids and bases in water. Generally, these calculate ions fall into two categories:

(1) calculating the value of K_a or K_b from the initial concentration of the acid or base and the measured pH of the solution (or some other data about the equilibrium composition of the solution) , and

(2) calculating equilibrium concentrations given K_a or K_b and initial concentrations.

(1) Calculating K_a or K_b from Initial Concentrations and Equilibrium Data:

In problems of this type, our goal is to obtain the equilibrium concentrations that are needed to evaluate the mass action expression in the K_a or K_b expression. (The reason we need equilibrium values, of course, is that the value of Q at equilibrium equals the equilibrium constant.) Usually, we are given the molar concentration of the acid or base as it would appear on the label of the bottle containing the solution. Also provided is information from which we can obtain the equilibrium concentrations. This might be the measured pH of the solution, which always provides the equilibrium concentration of H^+. Instead. we might be given the percentage ionization of the acid or base, which we can define as follows:

$$\text{Percentage ionization} = \frac{\text{amount ionized}}{\text{amount available}} \times 100\% \qquad (11)$$

Let's look at some examples that illustrate how to determine K_a and K_b from the kind of data mentioned.

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EX.5

Formic acid, $HCHO_2$, is a monoprotic acid. In a 0.1M solution of formic acid, the pH is 2.38 at 25 °C. Calculate the K_a and pK_a for formic acid at this temperature.

Analysis

In any problem like this we begin by writing the equation for the equiibrium so that we can set up the expression for K_a . We know the solute is an acid, and we know the general equation for the ionization of an acid, which we apply to the solute in question.

$$HCHO_2 = H^+ + CHO_2^-$$

$$K_a = \frac{[H^+][CHO_2^-]}{[HCHO_2]}$$

Keep in mind that the molar concentrations in the K_a expression all refer to equilibrium values. Also, in this problem the equilibrium concentration of H^+ must be identical to that of CHO_2^- because the ionization gives these ions in a 1 : 1 ratio. However, we do not have the values of either $[H^+]$ or $[CHO_2^-]$; all we have is the pH of the 0. 10 M solution. We must start by finding $[H^+]$ from pH. This gives us the equilibrium value of $[H^+]$. Then we'll put together a concentration table.

Solution

First we convert pH into $[H^+]$.

$[H^+] = -10^{-2.38} = 4.2 \times 10^{-3}M - 0.0042M$ (the form most useful next)

Because $[CHO_2^-] = [H^+]$, we now know that $[CHO_2^-] = 0.0042$ M.

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With these data we can now set up our concentration table.

	$HCHO_2$ =	H^+ +	CHO_2^-
Initial concentration(M)	**0.1**	**0.0**	**0.0**
Change in concentration(M)	**-0.0042**	**+0.0042**	**+0.0042**
Equilibrium concentration(M)	**0.1-0.0042**	**0.0042**	**0.0042**
	= 0.096		

The last row of data gives us the equilibrium concentrations that we now use to calculate K_a.

$$K_a = \frac{(4.2 \times 10^{-3})(4.2 \times 10^{-3})}{0.096}$$
$$= 1.8 \times 10^{-4}$$

Thus the acid ionization constant for formic acid is 1.8×10^{-4}. To find pK_a, we take the negative logarithm of K_a.

$$pK_a = -\log K_a$$
$$= -\log(1.8 \times 10^{-4}) \quad \text{then,} \quad pK_a = 3.74$$

EX.6

Methylamine, CH_3NH_2, is a weak base and one of several substances that give herring brine its pungent odor. In 0.1M CH_3NH_2, 6.3% of the base has under gone ionization. What is the K_b of methylamine?

Solution

The first step is to write the chemical equation for the equilibrium the equilibrium law. Because CH_3NH_2 is a weak base, we have:

$$CH_3NH_2(aq) + H_2O = CH_3NH_3^+(aq) + OH^-(aq)$$

$$K_b = \frac{[CH_3NH_3^+][OH^-]}{[CH_3NH_2]}$$

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The percentage ionization tells us that 6.3% of the CH_3NH_2 has reacted. Therefore, the moles per liter of the base that has ionized at equilibrium in this solution is:

Moles per liter of CH_3NH_2 ionized = 0.063 x 0.1 M = 0.0063 M

This represents the change in the concentration of CH_3NH_2, which we can now use to determine the concentrations of the other species at equilibrium. To do this. can set up the concentration table.

	H_2O +	CH_3NH_2 =	$CH_3NH_3^+$ +	OH^-
Initial concentration(M)		**0.1**	**0.0**	**0.0**
Change in concentration(M)		**-0.0063**	**+0.0063**	**+0.0063**
Equilibrium concentration(M)		**0.1-0.0063**	**0.0063**	**0.0063**
		= 0.094		

Now we can calculate K_b.

$$K_b = \frac{(0.0063)(0.0063)}{(0.094)}$$
$$= 4.2 \times 10^{-4}$$

and $\quad pK_b = -\log(4.2 \times 10^{-4}) = 3.38$

Calculating Equilibrium Concentrations from K_a (or K_b) and Initial Concentrations

Acid—base equilibrium problems of this type often present a confusing picture to students. Usually, the difficulty is getting started on the right foot. Therefore, before we begin, let's take an overview of the "landscape" to develop a strategy for selecting the correct attack on the problem.

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Almost any problem in which you are given K_a or K_b falls into one of three categories:

(1) the solution contains a weak acid as its only solute,

(2) the solution contains a weak base as its only solute, and

(3) the solution contains both a weak acid and its conjugate base.

<u>The following describes the approache take for each of these conditions.</u>

1. If the solution contains only a weak acid as the solute, then the problem must be solved using K_a. This means that the correct chemical equation for the problem is the ionization of the weak acid. It also means that if you are given the K_b for the acid's conjugate base, you will have to calculate the K_a in order to solve the problem.

2. If the solution contains only a weak base, the problem must be solved using K_b . The correct chemical equation is for the ionization of the weak base. If the statement of the problem provides you with the K_a for the base's conjugate acid, then you must calculate the K_b to solve the problem.

3. Solutions that contain two solutes, one a weak acid and the other its conjugate base, have some special properties that we will discuss later. To work problems for these kinds of mixtures, we can use either K_a or K_b—it doesn't matter which we use, we will obtain the same answers either way. However, if we elect to use K_a, then the chemical equation we use must be for the ionization of the acid. If we decide to use K_b then the correct chemical equation is for the ionization of the base.

When you begin the solution of a problem, decide which of the three conditions above applies. This will guide you in selecting the correct path to follow.

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Simplifications in Acid—Base Equilibrium Calculations

You learned that when the equilibrium constant is small it is frequently possible to make simplifying assumptions that greatly reduce the gebraic effort in obtaining equilibrium concentrations. For most acid—base equilibrium problems such simplifications are particularly useful.

Let's consider a solution of 1 M acetic acid, $HC_2H_3O_2$, for which $K_a = 1.8 \times 10^{-5}$. What is involved in determining the equilibrium concentrations for the solution?

To answer this question, we begin with the chemical equation for the equilibrium. Because the only solute in the solution is the weak acid, we must use K_a and therefore write the equation for the ionization of the acid. Let's use the simplified version. $HC_2H_3O_2 = H + C_2H_3O_2^-$

The equilibrium law is:

$$K_a = \frac{[H^+][C_2H_3O_2^-]}{[HC_2H_3O_2]} = 1.8 \times 10^{-5}$$

We will take the given concentration, 1.0 M, to be the initial concentration of $HC_2H_3O_2$ (i.e., the concentration of the acid before any ionization has occurred). This concentration will drop slightly as the acid ionizes and the ions form. If we let x be the amount of acetic acid that ionizes per liter, we can construct the following concentration table.

	CH_3COOH =	H^+ +	CH_3COO^-
Initial concentration(M)	**1.0**	**0.0**	**0.0**
Change in concentration(M)	**-x**	**+x**	**+x**
Equilibrium concentration(M)	**1.0-x**	**x**	**x**

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The equilibrium constant, 1.8×10^{-5}, is quite small, so we can anticipate that very little of the acetic acid will be ionized at equilibrium. This means x will be very small, so we make the approximation $(1 - x) \approx 1$. Substituting values into the K_a expression,

$$1.8 \times 10^{-5} = \frac{(x)(x)}{1.0}$$
$$= \frac{x^2}{1.0}$$

Solving for x gives x = 0.0042 M. We see that the value of x is indeed negligble compared to 1.0M (i.e.,if we subtract 0.0042M from 1.0M and round correctly, we obtain 1 M).

Notice that when we make this approximation, the initial concentration of the acid is used as if it were the equilibrium concentration. The approximation is valid when the equilibrium constant is small and the concentrations of the solutes are reasonably high — conditions that will apply to situations you will encounter in all

Let's look at some examples that illustrate typical acid—base equilibriurm problems.

EX.7

A solution of hydrazine, N_2H_4 , has a concentration of 0.25 M What is the pH of solution and what is the percentage ionization of the hydrazine? Hydrazine has K_b 1.7×10^{-6}

Solution

Let's write the chemical equation and the K_b expression.

$$N_2H_4 + H_2O = N_2H_5^+ + OH^-$$

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$$K_b = \frac{[N_2H_5^+][OH^-]}{[N_2H_4]}$$

The only sources of $N_2H_5^+$ and OH^- are the ionization of the N_2H_4 . so they will n in equal amounts. Let's call their equilibrium concentrations x.

$$[N_2H_5^+] = [OH^-] = x$$

Because K_b so small, $[N_2H_4] \approx 0.25M$ (In other words, we are using the initial ceritration as the equilibrium concentration; that's our approximation, which we expect to be valid.) Substituting into the K_b expression,

$$\frac{(x)(x)}{0.25} = 1.7 \times 10^{-6}$$

Solving for x gives: $x = 6.5 \times 10^{-4}$.This value represents the hydroxide ion concentration, from which we can calculate the pOH.

$$pOH = -\log(6.5 \times 10^{-4}) = 3.19$$

The pH of the solution can then be obtained from the relationship

$$pH + pOH = 14$$

Thus,

$$pH = 14.00 - 3.19 = 10.81$$

To calculate the percentage ionization, we need to know the amount of the base that has ionized. This value is also equal to x, because the amount of $N_2H_5^+$, that forms per liter equals the amount of N_2H_4 that ionizes. Therefore,

$$\text{Percentage ionization} = \frac{6.5 \times 10^{-4}}{0.25} \times 100\%$$

$$= 0.26\%$$

The base is 0.26% ionized.

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5. Solutions of Salts: Ions as Weak Acids and Bases

You saw that weak acids and bases are not limited to molecular substances. To prepare solutions of these ions, however, we cannot simply add them to water. Ions always come to us in compounds in which there is both a cation an anion. Therefore, to place NH_4^+ in water, we need a salt such as NH_4Cl, and to place $C_2H_3O_2^-$ in water, we need a salt such as $NaC_2H_3O_2$.

Because a salt contains two ions, the pH of its solution can potentially be affected by either the cation or the anion, or perhaps even by both. Therefore, we have to consider both ions as we assess the effect of the salt on the pH of the solution.

Cations as Acids

Conjugate Acids of Molecular Bases Are Weak Acids .

The conjugate of any molecular base is a cation. For example, NH_4^+ is the conjugate acid the molecular base, NH_3 . The equilibrium and K_a expression for NH_4^+ are

$$NH_4^+ (aq) = NH_3(aq) + H^+(aq)$$

Because the K_a values for ions are seldom tabulated, we would usually expect to calculate the K_a value using the relationship:

$$K_a \cdot K_b = K_w .$$

the K_b for NH_3 is given as 1.8×10^{-5}. Therefore, for NH_4^+

$$K_a = \frac{K_w}{K_b} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-5}} = 5.6 \times 10^{-10}$$

The hydrazinium ion, $N_2H_5^+$, is also a weak acid.

$$N_2H_5^+ (aq) = N_2H_4(aq) + H^+(aq)$$

$$K_a = \frac{[N_2H_4][H^+]}{[N_2H_5^+]}$$

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The tabulated K_b for N_2H_4 is 1.7×10^{-6}. From this, the calculated value of K_a for $N_2H_5^+$ equals 5.9×10^{-9}.

These examples illustrate that the conjugate acids of molecular bases tend to be weak acids that are capable of affecting the H of a solution. This leads to the following generalization.

1-Salts that contain cations of weak molecular bases can affect the pH of a solution. These cations are weak acids.

2-Salts that contain small ions with large charges, such as Al^{+3} and Cr^{3}, often produce solutions that are acidic.

3-The anion of a weak acid is a weak base and can influence the pH of a solution.

4-The anion of a strong acid is not basic and cannot influence the pH of a solution.

Predicting the Acid—Base Properties of a Salt

To decide whether any given salt will affect the pH of an aqueous solution, we examine each of its ions and see what it alone might do. There are three possibilities:

1. If only the cation of the salt is acidic, the solution will be acidic.

2. If only the anion of the salt is basic the solution will be basic.

3. If a salt has a cation that is acidic and an anion that is basic, the pH of the present). solution is determined by the relative strengths of the acid and base.

Let's work some examples to show how to use these generalizations.

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EX.8

Sodium hypochlorite, NaOC1, is an ingredient in many common bleaching and Predicting the Effect of a Salt disinfecting agents. Will a NaOCl solution be acidic, basic, or neutral?

Analysis

This is a salt, so it dissociates in water 100%.

$$NaOC1(s) \xrightarrow{H_2O} Na^+(aq) + OC1^-(aq)$$

We take each ion, in turn, and examine its effect of the acidity of the solution.

Solution

The Na^+ ion is of a Group IA metal, so it does not form an acidic hydrate. The cation, therefore, is neutral and is not acidic.

The OCl^- ion is the conjugate base of HOCl, which must be a weak acid (because it isn't on the list of strong acids you're expected to know). Therefore, OCl^- is a weak base and its presence should tend to make the solution basic.

Of the two ions in the salt, one is basic and the other is neutral. The answer the question, therefore, is that this salt solution will be basic.

EX.9

What is the pH of a 0.1 M solution of NaOCl? For HOCl, K_a 3.0 x 10^{-8}.

Analysis

Problems such as this are just like the other acid—base equilibrium problems you have learned to solve. We proceed as follows:

(1) We determine the nature of the solute—is it a weak acid, a weak base, or are both a weak acid and weak base present?

(2) We write the appropriate chemical equation and equilibrium law.

(3) We proceed with the solution.

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Solution

The solute is a salt, which is dissociated into the ions Na^+ and OCl^- .Only the latter can affect the pH. It is the conjugate base of HOCl, so for the purposes of problem solving, the active solute species is a weak base.

You learned that when the only solute is a weak base, we must use the K_b expression to solve the problem. Therefore, the next step is to write the appropriate chemical equation and the K_b expression.

$$OCl^- + H_2O \rightleftharpoons HOCl + OH^- \qquad K_b = \frac{[HOCl][OH^-]}{[OCl^-]}$$

The data provided in the problem is the K_a HOCl. But, we can easily calculate K_b because $K_a \cdot K_b = K_w$. Thus, for OCl^-,

$$K_b = \frac{1.0 \times 10^{-14}}{3.0 \times 10^{-8}} = 3.3 \times 10^{-7}$$

The only sources of HOCl and OH^- are the reaction of the OCl^-, so they will form in equal amounts. Let' s call their equilibrium concentrations x.

$$[HOCl] = [OH^-] = x$$

Because K_b is so small, $[OCl^-] \approx 0.10M$. (Once again, we are using the initial concentration as the equilibrium concentration; that's our approximation, which we expect to be valid.) Substituting into the K_b expression,

$$\frac{(x)(x)}{0.10} = 3.3 \times 10^{-7}$$

$$x = 1.8 \times 10^{-4}\ M$$

This value of x represents the 0H concentration, from which we can calculate the pOH and then the pH.

$$pOH = -\log(1.8 \times 1o^{-4})$$

$$pH = 14.00 - pOH = 14.00 - 3.74 = 10.26$$

The pH of this solution is 10.26.

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EX.10

What is the pH of a 0.2 M solution of hydrazinium chloride, N_2H_5Cl? Hydrazine N_2H_4,is a weak base with $K_b = 1.7 \times 10^{-6}$.

Solution

We begin with the chemical equation for the equilibrium and write the K_a expression.

$$N_2H_5^+ \rightleftharpoons H^+ + N_2H_4 \qquad K_a = \frac{[H^+][N_2H_4]}{[N_2H_5^+]} = \frac{1.0 \times 10^{-14}}{1.7 \times 10^{-6}} = 5.9 \times 10^{-9}$$

When the $N_2H_5^+$ reacts, equal amounts of H^+ and N_2H_4 are formed, so we let the equilibrium concentrations of each be equal to x. $[H^+] = [N_2H_4] = x$

We also assume that $[N_2H_5^+] \approx 0.20$ M and then substitute quantities into the mass action expression.

$$\frac{(x)(x)}{0.20} = 5.9 \times 10^{-9}$$

$$x = 3.4 \times 10^{-5}\ M$$

Since $x = [H^+]$, the pH of the solution is

$$pH = -\log(3.4 \times 10^{-5}) = 4.47$$

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Solutions Made with the Salt of a Weak Acid and a Weak Base

Many salts involve ions both of which are capable of affecting the pH of the solution. Whether or not the salt has a net effect on the pH now depends on the relative strengths of its ions in functioning one as an acid and the other base. If they are matched in their respective strengths, the salt has no effect on pH. In ammonium acetate, for example, the ammonium ion is an acidic cation and the acetate ion is a basic anion. However, the K_a of NH_4^+ is 5.6×10^{-10} and the K_b of $C_2H_3O_2^-$ just happens to be the same, 5.6×10^{-10}. The cation tends to produce H^+ ions to the same extent that the anion tends to produce OH^-. So in aqueous ammonium acetate, $[H^+] = [OH^-]$, and the solution has a pH of 7.

Consider, now, ammonium formate, NH_4CHO_2 . The formate ion, CHO_2^-, is the conjugate base of the weak acid, formic acid, so it is a Bronsted base Its K_b 5.6×10^{-11}. Comparing this value to the (slightly larger) K_a of the ammonium ion, 5.6×10^{-10}, we see that NH_4^+ is slightly stronger as an acid the formate ion can be as a base. So a solution of ammonium formate will be slightly acidic. We are not concerned here about calculating a pH, one predicting whether the solution is acidic,basic,or neutral.

EX.11

Will an aqueous solution that is 0.20M NH_4F be acidic, basic, or neutral?

Solution

The K_a of NH_4^+ (calculated from the K_b for NH_3) is 5.6×10^{-10} . Similarly, the K_b of F^- is 1.5×10^{-11} (calculated from the K_a for HF, 6.8×10^{-4}). Comparing the two equilibrium constants, we see that the acid (NH_4^+) is stronger the base (F^-), so the solution is slightly acidic.

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